Plant Disease Detection using Deep Learning & Fertilizer Recommendation

Project Overview

This project utilizes deep learning techniques to detect plant diseases from images. The model is based on **ResNet** architecture and is trained on a dataset containing images of different plant diseases. The web application allows users to upload images, detect diseases, and get fertilizer recommendations.

System Requirements

- **Operating System:** Windows/Linux/Mac
- **Python Version:** 3.8+
- **Libraries Required:**
    - TensorFlow/Keras
    - PyTorch (if using PyTorch version of model)
    - OpenCV
    - NumPy
    - Pandas
    - Matplotlib
    - Django (for web interface)
    - SQLite (for database management)

Installation & Setup Instructions

Step 1: Unzip the Project

- Extract the provided ZIP file using WinRAR/7-Zip.
- Navigate to the extracted folder.

Step 2: Create a Virtual Environment

```
python -m venv plant_env
```

- Activate the environment:
    - **Windows:** `plant_env\Scripts\activate`
    - **Linux/Mac:** `source plant_env/bin/activate`

Step 3: Install Dependencies

```
pip install -r requirements.txt
```

Step 4: Set Up the Database

```
python manage.py migrate
```

Step 5: Run the Server

```
python manage.py runserver
```

- Open the browser and go to `http://127.0.0.1:8000/`

Default Login Credentials

- **Username:** `Admin`
- **Password:** `admin123`

Dataset Details

- The dataset consists of **38 plant disease categories** with a total of **70,302 images**.

Testing the Model

1. **Navigate to the Test Folder**
   - Open the extracted folder and locate the test folder.
2. **Select an Image for Testing**
   - Inside the test folder, you will find sample images for testing.
   - Choose any image you want to test.
3. **Upload the Image in the Web Interface**
   - Open the application.
   - Click on the **Upload Image** button.
   - Select an image from the test folder.
   - Click the **Detect** button to view the results.

Model Architecture (ResNet)

- **Convolutional Layers**: Extract features from images.
- **Residual Blocks**: Enhance gradient flow.
- **Batch Normalization & ReLU**: Improve stability.
- **Fully Connected Layer**: Classifies images into **38 categories**.

Web Interface Features

- **Index Page:** User Registration & Login.
- **Home Page:** Upload image & Get disease prediction.
- **Performance Page:** Shows model accuracy & results.

Usage Guide

1. **Login/Register** on the web application.
2. **Upload an Image** of a plant leaf.
3. Click **'Detect'** to get disease name and fertilizer recommendation.
4. View **Performance Metrics** on the dedicated page.

Expected Output & Screenshots

- **Validation Accuracy Graph**
  - `validation_accuracy.png`
- **Sample Detection Output**
  - Uploaded Image
  - Predicted Disease Name
  - Suggested Fertilizer & Treatment Plan

Conclusion

This project efficiently detects plant diseases using deep learning and provides recommendations to farmers and agricultural professionals.

Dataset: New Plant Diseases Dataset

Hardware Requirements for Training the Model

To train the deep learning model for **Plant Disease Detection**, the following hardware requirements are recommended: Minimum Requirements

1. Minimum Requirements (For small-scale training or testing)

- **Processor:** Intel Core i5 (10th Gen or later) / AMD Ryzen 5
- **RAM:** 8 GB
- **GPU:** NVIDIA GTX 1050 Ti (4GB VRAM) or higher
- **Storage:** 50 GB free space (for dataset & model checkpoints)
- **Operating System:** Windows 10/11, Ubuntu 20.04+
- **Other:** Python 3.8+, TensorFlow/PyTorch, CUDA installed

2. Recommended Requirements (For faster training & larger datasets)

- **Processor:** Intel Core i7/i9 (12th Gen or later) / AMD Ryzen 7/9
- **RAM:** 16 GB or more
- **GPU:** NVIDIA RTX 3060 (12GB VRAM) or higher (RTX 3090 / A100 for large datasets)
- **Storage:** 200 GB SSD (for better speed)
- **Operating System:** Ubuntu 20.04+ (preferred) / Windows 11
- **Other:** CUDA-enabled GPU, cuDNN, TensorFlow/PyTorch optimized

3. High-Performance Setup (For large-scale training & deep models)

- **Processor:** AMD Threadripper / Intel Xeon (multi-core)
- **RAM:** 32GB - 128GB (for handling large batch sizes)
- **GPU:** NVIDIA RTX 4090 / A100 / Tesla V100 (for best performance)
- **Storage:** 1TB NVMe SSD (for dataset, logs, and model checkpoints)
- **Operating System:** Linux-based (Ubuntu 22.04 recommended)
- **Other:** High-speed internet, multi-GPU setup (for distributed training)